

Myhill Nerode Theorem and Equivalence Classes $\approx_L$

(a) $\approx_L$ is an equivalence relation

Proof. To show that $\approx_L$ is an equivalence relation, we prove that it satisfies symmetry, reflexivity, and transitivity:

1. **Reflexivity:** We claim $x \approx_L x$; assume not, then $\exists z \in \Sigma^* : xz \in L \wedge xz \notin L$, which is a fallacy.
2. **Symmetry:** Let $x \approx_L y$, we claim $y \approx_L x$. As $x \approx_L y$:

$$\begin{aligned}
 &\Rightarrow \neg \exists z : xz \in L \wedge yz \notin L \\
 &\Rightarrow \forall z : xz \in L \Leftrightarrow yz \in L \\
 &\Rightarrow \forall z : yz \in L \Leftrightarrow xz \in L \\
 &\Rightarrow \neg \exists z : yz \in L \wedge xz \notin L \\
 &\Rightarrow y \approx_L x
 \end{aligned}$$

3. **Transitivity:** Let $x \approx_L y$, $y \approx_L z$. We claim $x \approx_L z$. As both statements hold simultaneously:

$$\begin{aligned}
 &\Rightarrow (\neg \exists k : xk \in L \wedge yk \notin L) \wedge (\neg \exists l : yl \in L \wedge zl \notin L) \\
 &\Rightarrow (\forall k : xk \in L \Leftrightarrow yk \in L) \wedge (\forall l : yl \in L \Leftrightarrow zl \in L) \\
 &\Rightarrow \forall k, l : (xk \in L \Leftrightarrow yk \in L) \wedge (yl \in L \Leftrightarrow zl \in L)
 \end{aligned}$$

Now set $k = l$ in the above predicate to obtain:

$$\Rightarrow \forall k : xk \in L \Leftrightarrow yk \in L \Leftrightarrow zk \in L$$

Thus we have established the equivalence of all three set memberships. So we can rephrase the predicate as:

$$\begin{aligned}
 &\Rightarrow \forall k : xk \in L \Leftrightarrow zk \in L \\
 &\Rightarrow \neg \exists k : xk \in L \wedge zk \notin L \\
 &\Rightarrow x \approx_L z
 \end{aligned}$$

This thus proves $\approx_L$ is an equivalence relation. □

(b) Distinguishable strings reach distinct states

Proof. Let $x, y \in \Sigma^*$ be such that they are distinguishable for a language L , i.e. $x \neq y$. Let the DFA M decide L . We claim that if M , starting from state q_0 , lands on state p while reading x and M lands on state q on reading y , then $p \neq q$.

Assume not, i.e. $p = q$. We know $x \neq y$, i.e. $\exists l \in \Sigma^*$ such that $xl \in L \wedge yl \notin L$. Thus consider the inputs xl and yl to M ; we know $\hat{\delta}(q_0, x) = \hat{\delta}(q_0, y) = p$, and as M is a DFA, $\hat{\delta}(p, l) = p^*$ (say). Hence $\hat{\delta}(q_0, xl) = \hat{\delta}(q_0, yl) = p^*$. Hence xl is accepted by the DFA M if and only if yl is accepted by the DFA M . But we know that xl is accepted and yl is not; contradiction. Hence $p \neq q$. □

(c) A distinguishing set of size k forces $\geq k$ states

Proof. Let L be a language such that there exists a distinguishing set of strings $S_L = \{x_1, x_2, \dots, x_k\}$. Let $x_i, x_j \in S$, $i \neq j$. We know via part (b) that any DFA M deciding L on inputs x_i and x_j must go to distinct states p_i and p_j respectively, as the strings are distinguishable. This must be true for any pair x_i, x_j , thus we must have at least one distinct state for each $x_i \in S_L \Rightarrow$ at least k states in the DFA M ; one for each x_i . $\square$

(d)(i) Infinitely many classes $\Rightarrow L$ is not regular

Proof. If Σ^* has infinite equivalence classes with respect to $\approx_L$ for some language L , then we claim that there does not exist any DFA M solving it, thus proving it is not regular.

Assume not, i.e. such a DFA M exists. Let I be the indexing set for the equivalence classes E_i ; i.e. $\bigcup_{i \in I} E_i = \Sigma^*$. Consider the set

$$S_L = \{x_\alpha \mid x_\alpha \text{ is the lexicographic smallest word in } E_\alpha, \alpha \in I\}.$$

First note x_α always exists by the well-ordering principle applied to Σ^* as it is a countable set.

Now, as $\forall i, j \in I, i \neq j \rightarrow x_i \neq x_j$, because x_i and x_j belong to different equivalence classes w.r.t. $\approx_L$. Hence each pair of strings in S_L is distinguishable from each other, making S_L a set of distinguishing strings. By part (c) we know that we must thus have a unique state q_i for each $x_i \in S_L$. But as $|S_L| = |I| = |\mathbb{N}|$, as there are infinite equivalence classes, there must be thus infinite states corresponding to each $x_i \in S_L$, and thus as M is a DFA and has only finite states, M cannot decide L and we are done. $\square$

(d)(ii) Finitely many classes $\Rightarrow L$ is regular

Proof. Let L be a given language with the number of equivalence classes $= T \in \mathbb{N}$. We claim that the following DFA M constructed below decides L :

1. For each equivalence class $E_i, i \in [T]$, construct a state q_i .
2. Define the transition function $\delta(q_i, a) = q_j$ if and only if $\forall x \in E_i, xa \in E_j$.
3. Set the start state as the state q_{start} , the state which has ε in its equivalence class, and make a state accepting if and only if $E_i \cap L \neq \phi$.

First note that $\delta : Q \times \Sigma \rightarrow Q$ as defined above is well-defined: Let $x, x' \in E_i$; we prove the classes of xa and $x'a$ are equal $\forall a \in \Sigma$. We know $x \approx_L x'$, thus $\forall z \in \Sigma^* : (xz \in L \Leftrightarrow x'z \in L)$. Let $z = aw, a \in \Sigma, w \in \Sigma^*$. Then by applying $z = aw$ in the predicate we obtain $\forall z = aw \in \Sigma^* : (xaw \in L \Leftrightarrow x'aw \in L) \Rightarrow \forall w \in \Sigma^* : (xaw \in L \Leftrightarrow x'aw \in L)$. Thus $xa \approx_L x'a$, and thus their equivalence classes are the same.

Now, we claim that M decides L . We first handle the case of deciding ε . We know that $q_{start} \in F \Leftrightarrow E_{start} \cap L \neq \phi$. If $q_{start} \notin F$ there is nothing to prove. If it is, let $x \in E_{start} \cap L \neq \phi$. Then $x \approx_L \varepsilon$. Thus $\forall z \in \Sigma^* : xz \in L \Leftrightarrow \varepsilon z = z \in L$. Thus by setting $z = \varepsilon$ we obtain $\varepsilon \in L$ as we know $x \in L$.

Let x be an input of non-zero length to M . Let $q = \hat{\delta}(q_{start}, x)$. We know that by the construction of δ , after first reading a string y and then a symbol a we move to the state corresponding to the equivalence class of ya , and thus we know that we must be in the equivalence class of x . If $q \notin F$ there is nothing to prove (as we know for sure then that $x \notin L$), so assume $q \in F$. We prove $x \in L$. Let x' witness $q \in F$, i.e. $x' \in E_q \cap L$. Then we know $x \approx_L x' \Rightarrow \forall z : xz \in L \Leftrightarrow x'z \in L$, and thus if we set $z = \varepsilon$ we obtain $x \in L$, since we know $x' \in L$, and hence we are done. $\square$